

Optimization Summary — web_b5258d7e4ad0

Run: web_b5258d7e4ad0_optimize_20260914T144523Z

Baseline vs Final

Metric	Baseline	Final	Change
Worst slack (ns)	-1.41	-0.71	+49.6% ↑ (better)
Max frequency (MHz)	343.6	463.0	+34.7% ↑ (better)
Cell area (μm ²)	61804	62802	+1.6% ↑ (worse)
Flip-flops	1952	2214	+13.4% ↑ (worse)
Total cells	48070	48230	+0.3% ↑
Total power (μW)	181000	167000	-7.7% ↓ (better)
Timing constraint met	False	False	

Accepted Rounds

- **Round 1** — mac_unit (clock clk_a): a structural change to mac_unit (see round_1/plan.txt if present for the planner's exact reasoning; not recorded for this run) Cost/effect: +1 cycle(s) of latency; 2116 flip-flop(s), 62519 μm² cell area; net weighted improvement +5.0%. Verified equivalent by SymbiYosys (depth 20), unbounded proof.
- **Round 2** — bm_dot4 (clock clk_c): a structural change to bm_dot4 (see round_2/plan.txt if present for the planner's exact reasoning; not recorded for this run) Cost/effect: +1 cycle(s) of latency; 2214 flip-flop(s), 62802 μm² cell area; net weighted improvement +37.7%. Verified equivalent by SymbiYosys (depth 20), unbounded proof.

Where It Stopped

The run stopped after round 3 (bm_fir6): All 2 candidate(s) could not be verified in time (the equivalence checker timed out).